```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

```
df=pd.read_csv('/content/Customer-Churn-analysis/Customer Churn.csv')
df
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetServ
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	D
1	5575-GNVDE	Male	0	No	No	34	Yes	No	D
2	3668-QPYBK	Male	0	No	No	2	Yes	No	D
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	D
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber c
...	...	...	...	...	...	...	...	...	
7038	6840-RESVB	Male	0	Yes	Yes	24	Yes	Yes	D
7039	2234-XADUH	Female	0	Yes	Yes	72	Yes	Yes	Fiber c
7040	4801-JZAZL	Female	0	Yes	Yes	11	No	No phone service	D
7041	8361-LTMKD	Male	1	Yes	No	4	Yes	Yes	Fiber c
7042	3186-AJIEK	Male	0	No	No	66	Yes	No	Fiber c

7043 rows × 21 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

replacing blank values(considered as object datatype) into 0 and changing datatype

```
df['TotalCharges']=df['TotalCharges'].replace(" " , "0")
df['TotalCharges']=df['TotalCharges'].astype("float")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
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Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   float64
 20  Churn             7043 non-null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1+ MB
```

```
df.isnull().sum()
```

	0
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

to check if there is duplicated row or not

```
df.duplicated()
```

	0
0	False
1	False
2	False
3	False
4	False
...	...
7038	False
7039	False
7040	False
7041	False
7042	False

7043 rows × 1 columns

dtype: bool

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df['customerID'].duplicated().sum()
```

```
np.int64(0)
```

converting the seniorcitizen data into yes/no value >>> easier to understand

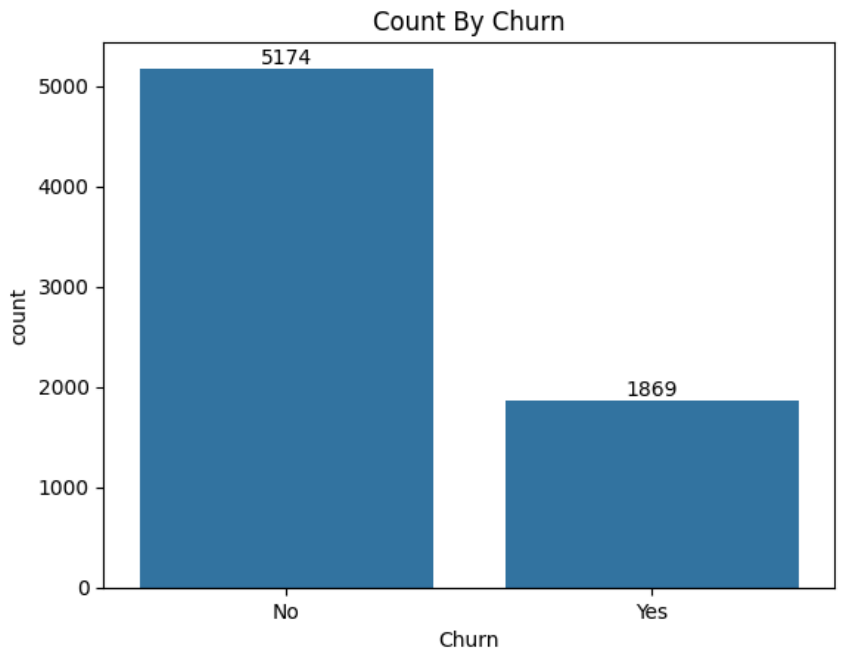
```
def conv(value):
  if value==1:
    return "yes"
  else:
      return "no"

df["SeniorCitizen"]=df["SeniorCitizen"].apply(conv)
```

```
df.head()
```

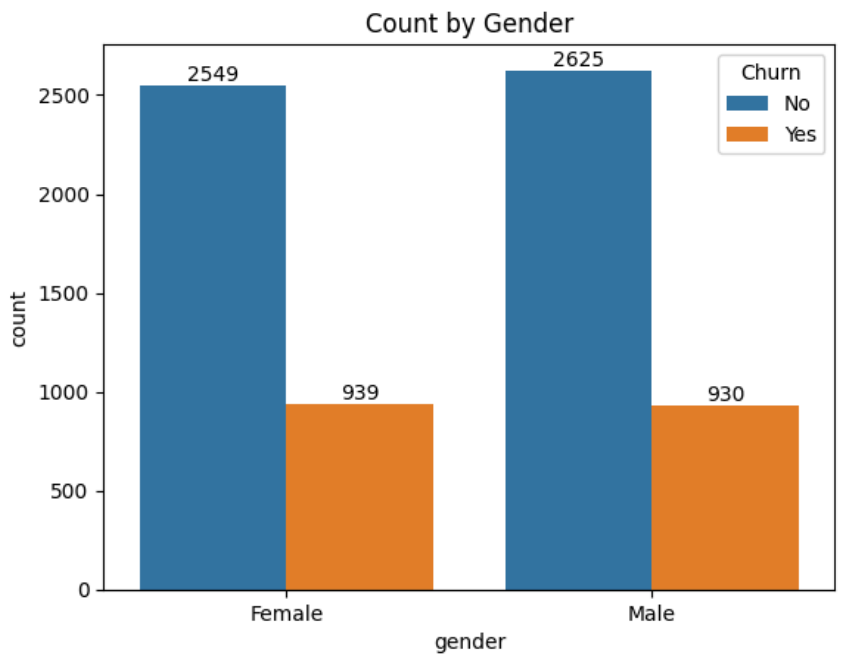
	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService
0	7590-VHVEG	Female	no	Yes	No	1	No	No phone service	DSL
1	5575-GNVDE	Male	no	No	No	34	Yes	No	DSL
2	3668-QPYBK	Male	no	No	No	2	Yes	No	DSL
3	7795-CFOCW	Male	no	No	No	45	No	No phone service	DSL
4	9237-HQITU	Female	no	No	No	2	Yes	No	Fiber optic

5 rows × 21 columns

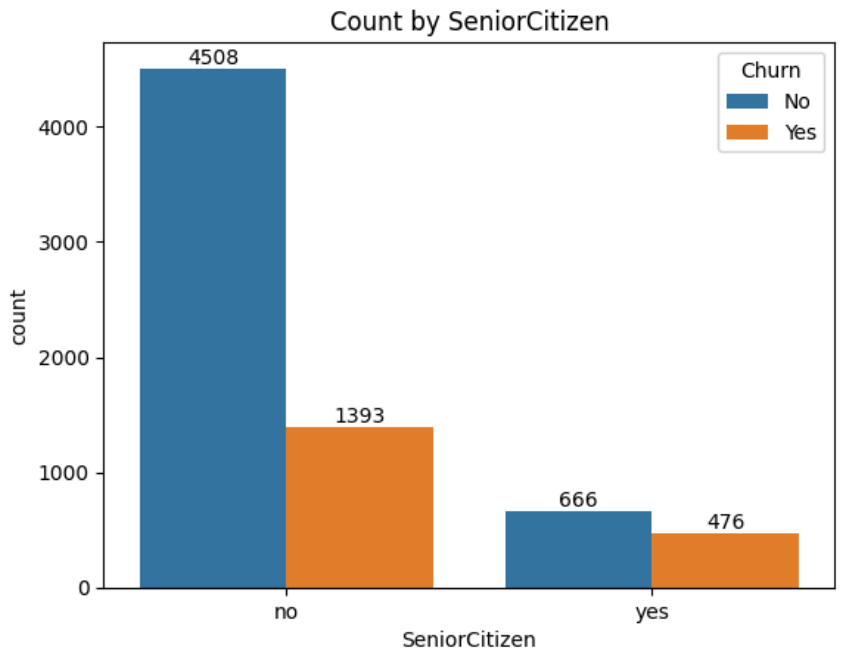
```
a=sns.countplot(x='Churn',data=df)
a.bar_label(a.containers[0])
plt.title("Count By Churn")
plt.show()
```



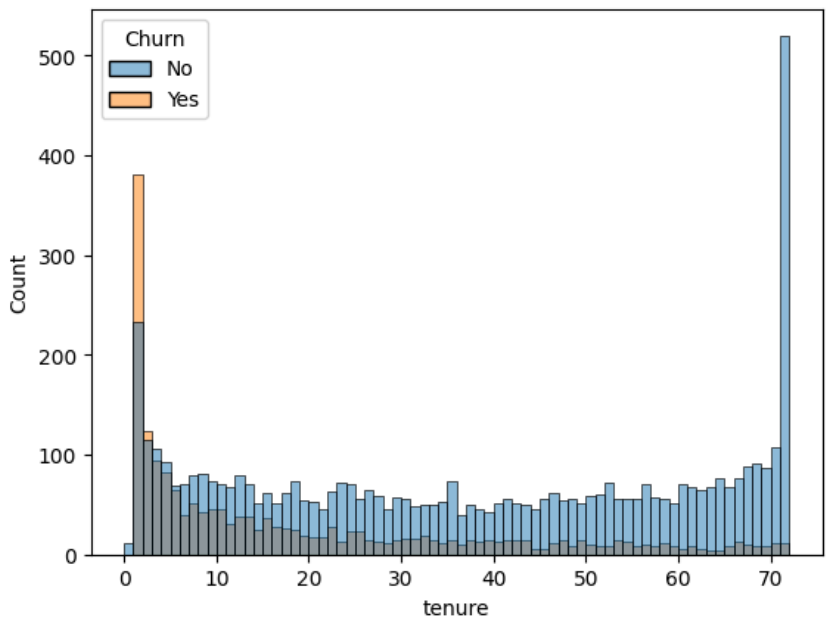

```
a=sns.countplot(x='gender',data=df,hue="Churn")
a.bar_label(a.containers[0])
a.bar_label(a.containers[1])
plt.title("Count by Gender")
plt.show()
```




```
a=sns.countplot(x='SeniorCitizen',data=df,hue="Churn")
a.bar_label(a.containers[0])
a.bar_label(a.containers[1])
plt.title("Count by SeniorCitizen")
plt.show()
```

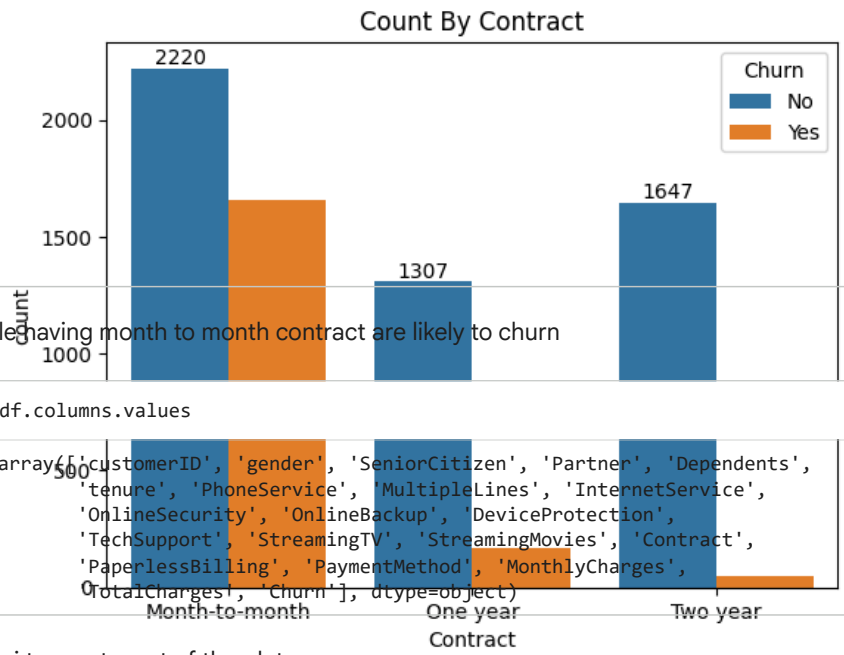



```
sns.histplot(x="tenure", data=df,bins=72,hue="Churn")
plt.show()
```



people who used the services for 1 or 2 months have churned while those who used it for long term have not left

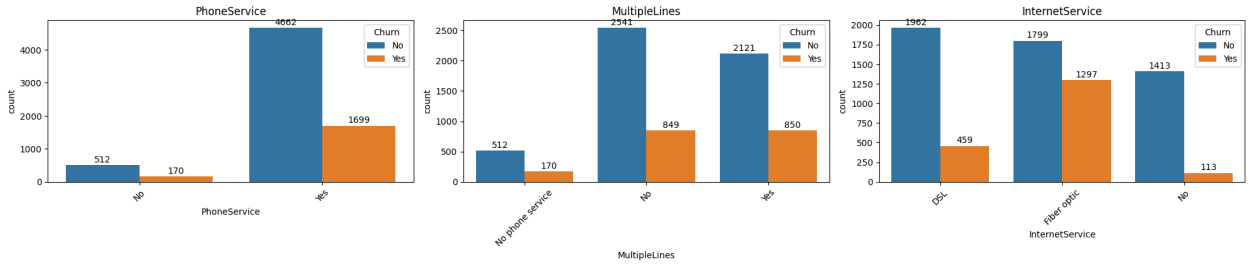
```
a=sns.countplot(x='Contract',data=df,hue="Churn")
a.bar_label(a.containers[0])
plt.title("Count By Contract")
plt.show()
```



people having month to month contract are likely to churn

```
df.columns.values
```

```
array(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
       'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',
       'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',
       'TotalCharges', 'Churn'], dtype=object)
```

used ai to create rest of the plots.

```
# Select categorical columns (excluding customerID & Churn)
cat_cols = [

    'PhoneService', 'MultipleLines', 'InternetService',
    'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
    'TechSupport', 'StreamingTV', 'StreamingMovies',
    'Contract', 'PaperlessBilling', 'PaymentMethod'
]

# Set subplot size
plt.figure(figsize=(20, 25))

for i, col in enumerate(cat_cols):
    plt.subplot(6, 3, i+1)
    ax = sns.countplot(x=col, data=df, hue="Churn")
    plt.title(col)
    plt.xticks(rotation=45)

    # Add labels
    for container in ax.containers:
        ax.bar_label(container, padding=1)

plt.tight_layout()
plt.show()
```